Ch 5 Section 1 Review

$$P(\underline{x}, D) = \sum_{|\alpha| \le m} a^\alpha D^\alpha$$ Linear operator

$$P_m(\underline{x}, D) = \sum_{|\alpha| = m} a^\alpha(\underline{x}) D^\alpha$$ Principal part of Linear operator

$$P_m(\underline{x}, \underline{\xi}) = \sum_{|\alpha| = m} a^\alpha(\underline{x}) \underline{\xi}^\alpha$$ principal symbol of P

<u>Def'n</u> A <u>characteristic</u>, $\underline{\xi} \neq \underline{0}$, is a direction (not concerned with magnitude or sign) for linear operator, $P(\underline{x}, D)$, if

$$P_m(\underline{x}, \underline{\xi}) = 0$$

<u>Idea</u>

Information of sol'n to PDE is spread ~~in~~ through a surface where $\underline{\xi}$ is normal to surface.

Ch 5 section 7

2nd-order linear PDE:

$$a(x,y)\,u_{xx} + b(x,y)\,u_{xy} + c(x,y)\,u_{yy}$$

↑ZT puts a 2 here.

$$+ d(x,y)\,u_x + e(x,y)\,u_y + f(x,y)\,u = g(x,y)$$

How to classify?

Important part is highest-order terms.

Characteristic eq'n $\quad p_2(\underline{x},\underline{\xi}) = 0$

$$a(x,y)(\xi_x)^2 + b(x,y)\,\xi_x\xi_y + c(x,y)(\xi_y)^2 = 0$$

Divide through by ξ_y^2 and solve for ξ_x/ξ_y

$$a(x,y)\left(\frac{\xi_x}{\xi_y}\right)^2 + b(x,y)\frac{\xi_x}{\xi_y} + c(x,y) = 0$$

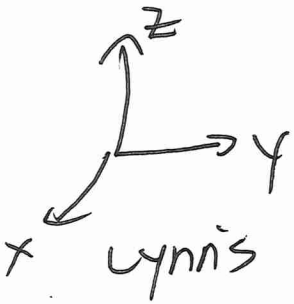
$$\frac{\xi_x}{\xi_y} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Consider the discriminant

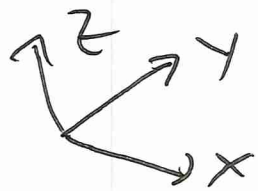
$$\Delta = b^2 - 4ac$$

$\uparrow$ Z&T has a 1 here

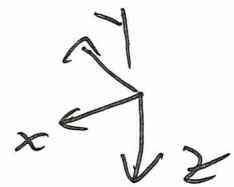
Classify PDE's based on fact that $\Delta > 0$, $\Delta = 0$, $\Delta < 0$?



Lynn's



Jacob



Justin

If we are going to classify PDE's, the classification should not depend on the coord. system.

If we change indep. variables $(x,y) \leftrightarrow (\xi,\eta)$

$\xi(x,y) \qquad \eta(x,y)$

change PDE $\quad a u_{xx} + b u_{xy} + c u_{yy} + \cdots = 0$

$$A(\xi,\eta)\, u_{\xi\xi} + B(\xi,\eta)\, u_{\xi\eta} + C(\xi,\eta)\, u_{\eta\eta} + \cdots = 0$$

may not be the same

Then $\Delta' = B^2 - 4AC$

(HW) $\Delta' = \Delta J^2$

where J is the Jacobian $\quad J = \begin{vmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \xi}{\partial y} \\ \frac{\partial \eta}{\partial x} & \frac{\partial \eta}{\partial y} \end{vmatrix}$

So the sign of the discriminant, Δ, does not change when changing coord! so this is good! Using sign of Δ to classify PDE is (physically) meaningful.

What happens if $J = 0$ or ∞?

Bad coord transformation

$\xi = x \quad \eta = 2x \qquad J = \begin{vmatrix} 1 & 0 \\ 2 & 0 \end{vmatrix} = 0$

can't go back & forth

$(x,y) \Leftrightarrow (\xi,\eta)$

$x = \xi \qquad y = ?$

Aside: $J = \dfrac{V_{\xi,\eta}}{V_{x,y}}$? $\quad V = \text{volume}$

~~Clasi~~

<u>classification</u>

locally

$\Delta > 0$ the eq'n is <u>hyperbolic</u>

$\Delta = 0$ " " " <u>parabolic</u>

$\Delta < 0$ " " " <u>elliptic</u>

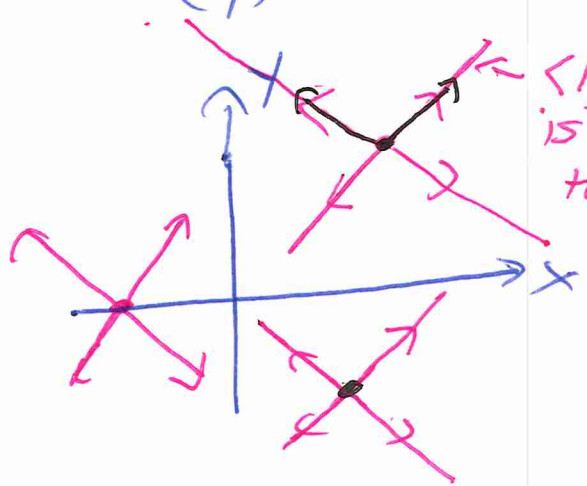
ex) $u_{xx} - u_{yy} + \underbrace{\ldots\ldots}_{\text{lower-order terms}} = 0$ <u>wave eqn</u>

$a = 1 \qquad b = 0 \qquad c = -1$

$\Delta = b^2 - 4ac = 0 - (4)(1)(-1) = 4 > 0$ <u>hyperbolic</u>

<u>characteristic eqn</u>

$$(\xi_x)^2 - (\xi_y)^2 = 0 \qquad \underline{\xi} = \boxed{\begin{matrix} \langle 1, 1 \rangle \\ \langle 1, -1 \rangle \end{matrix}}, \qquad \begin{matrix} \langle 2, 2 \rangle & \cancel{\langle 0, 0 \rangle} \\ \langle -1, 1 \rangle & \langle 300, -300 \rangle \end{matrix}$$



$\langle 1, -1 \rangle$ is $\perp$ to "surface"

At any pt in $\mathbb{R}^2$, information is propagated along 45° lines.

ex) $u_{xx} - u_t = 0$ heat eq'n

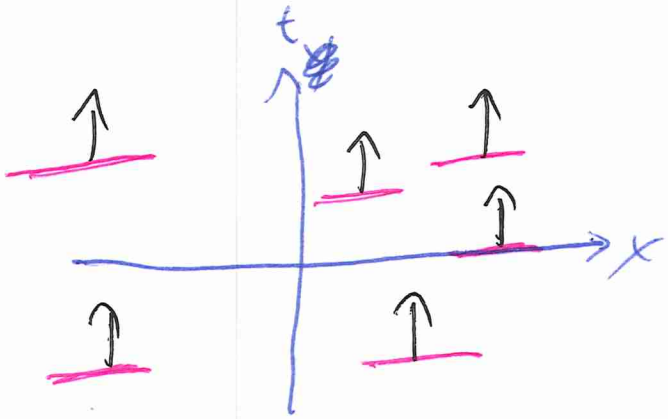
$a = 1 \quad b = 0 \quad c = 0$

$\Delta = b^2 - 4ac = 0^2 - 4(1)(0) = 0$ parabolic

charact. eq'n

$(\xi_x)^2 = 0$

charact. directions $\boxed{\langle 0, 1 \rangle}$ $\langle 0, -1 \rangle$



ex) Laplace eq'n $u_{xx} + u_{yy} = 0$

$a = 1 \quad b = 0 \quad c = 1$

$\Delta = b^2 - 4ac = 0 - 4(1)(1) = -4 < 0$ elliptic

charact. eq'n: $(\xi_x)^2 + (\xi_y)^2 = 0$ charact. directions? Are no charact. directions

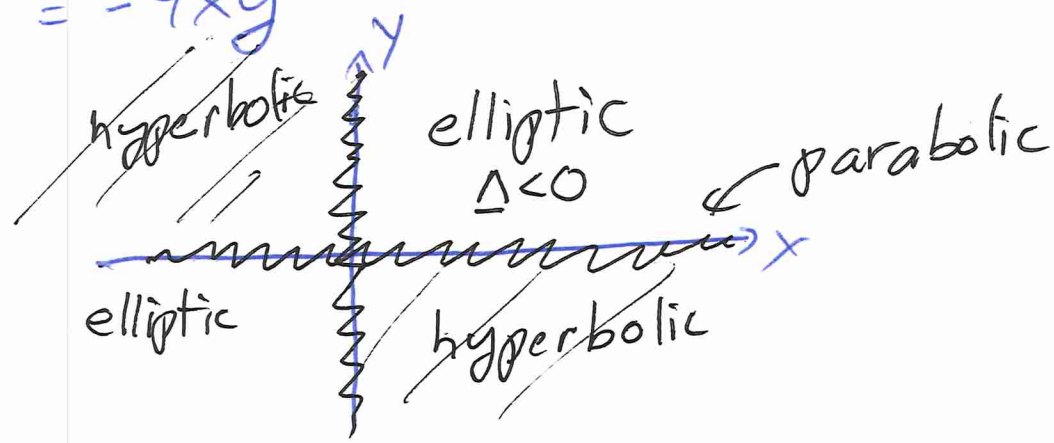
There's no preferred direction for information to travel.

ex) Describe the region in the (xy) plane where the PDE is hyperbolic ($\Delta > 0$), parabolic ($\Delta = 0$), or elliptic ($\Delta < 0$).

$$x u_{xx} + y u_{yy} + \sin(y) u_x - e^{x+y} u = x^2$$

$a = x \qquad b = 0 \qquad c = y$

$\Delta = b^2 - 4ac = 0^2 - 4xy = -4xy$

$-4xy = 0$

$x = 0$ or $y = 0$